

Lecture 15

Interpolation of Spatial Data II

STAT 8020 Statistical Methods II

Review: Spatial
Interpolation

Parameter estimation

A Case Study of
Paraná State
Precipitation Data

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If

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

Then

$$[Y_1 | Y_2 = y_2] \sim N(\mu_{1|2}, \Sigma_{1|2})$$

where

$$\mu_{1|2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (y_2 - \mu_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

Main Idea

Observed values provide information about unobserved values through the covariance structure

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If $\{Y(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}}$ follows a GP, then

$$\begin{pmatrix} Y_0 \\ \mathbf{Y} \end{pmatrix} \sim N \left(\begin{pmatrix} m_0 \\ \mathbf{m} \end{pmatrix}, \begin{pmatrix} \sigma_0^2 & k^T \\ k & \Sigma \end{pmatrix} \right)$$

We have

$$[Y_0 | \mathbf{Y} = \mathbf{y}] \sim N(m_{Y_0 | \mathbf{Y} = \mathbf{y}}, \sigma_{Y_0 | \mathbf{Y} = \mathbf{y}}^2)$$

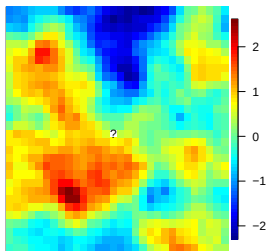
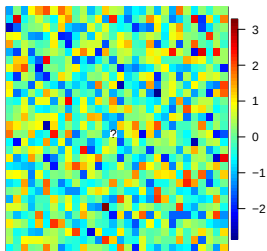
where

$$\begin{aligned} m_{Y_0 | \mathbf{Y} = \mathbf{y}} &= m_0 + k^T \Sigma^{-1} (\mathbf{y} - \mathbf{m}) \\ \sigma_{Y_0 | \mathbf{Y} = \mathbf{y}}^2 &= \sigma_0^2 - k^T \Sigma^{-1} k \end{aligned}$$

Next, we are going to revisit our toy examples

Toy Examples Revisited

For simplicity, we assume $m(s) = 0$ for $s \in \mathcal{S}$, the spatial covariance only depends on distance



$$m_{Y_0|Y=y} = 0 + k^T \Sigma^{-1} (y - 0), \quad \sigma_{Y_0|Y=y}^2 = \sigma_0^2 - k^T \Sigma^{-1} k$$

Spatial uncorrelated field:

- $m_{Y_0|Y} = 0$
- $\sigma_{Y_0|Y=y}^2 = \sigma_0^2$

Spatial correlated field:

- $m_{Y_0|Y} = k^T \Sigma^{-1} y$
- $\sigma_{Y_0|Y=y}^2 = \sigma_0^2 - k^T \Sigma^{-1} k$

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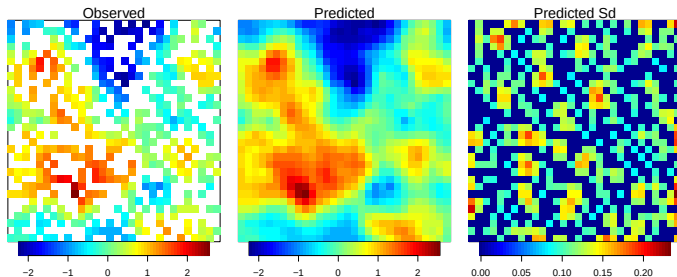
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Interpolating Multiple Points in Space

For prediction at multiple locations, the Gaussian conditional distribution still applies: $[Y_0|Y = y] \sim N(m_{Y_0|Y=y}, \Sigma_{Y_0|Y=y})$, where

$$m_{Y_0|Y=y} = m_0 + k^T \Sigma^{-1} (y - m)$$

$$\Sigma_{Y_0|Y=y} = \Sigma_0 - k^T \Sigma^{-1} k$$



Kriging produces both a prediction map and an uncertainty map

If $\{Y(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}}$ follows a GP, then

$$\begin{pmatrix} \mathbf{Y}_0 \\ \mathbf{Y} \end{pmatrix} \sim N \left(\begin{pmatrix} \mathbf{m}_0 \\ \mathbf{m} \end{pmatrix}, \begin{pmatrix} \Sigma_0 & \mathbf{k}^T \\ \mathbf{k} & \Sigma \end{pmatrix} \right)$$

We have

$$[\mathbf{Y}_0 | \mathbf{Y} = \mathbf{y}] \sim N(\mathbf{m}_{\mathbf{Y}_0 | \mathbf{Y} = \mathbf{y}}, \Sigma_{\mathbf{Y}_0 | \mathbf{Y} = \mathbf{y}})$$

where

$$\mathbf{m}_{\mathbf{Y}_0 | \mathbf{Y} = \mathbf{y}} = \mathbf{m}_0 + \mathbf{k}^T \Sigma^{-1} (\mathbf{y} - \mathbf{m})$$

$$\Sigma_{\mathbf{Y}_0 | \mathbf{Y} = \mathbf{y}} = \Sigma_0 - \mathbf{k}^T \Sigma^{-1} \mathbf{k}$$

Question: what if we don't know $m(\mathbf{s}; \boldsymbol{\beta}), c(h; \boldsymbol{\theta})$?

⇒ The mean and covariance functions are typically unknown and must be estimated from the data $\mathbf{y}$. This will be covered in this module

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Recap: Gaussian Process Model

Assume the observed data

$$\{y(s_i)\}_{i=1}^n$$

are one partial realization of a spatial stochastic process

$$\{Y(s) : s \in S\}.$$

A Gaussian process model is written as

$$Y(s) = m(s) + \epsilon(s), \quad \epsilon(s) \sim GP(0, K).$$

Under stationarity and isotropy,

$$K(s_i, s_j) = C(\|s_i - s_j\|) = C(h).$$

Modeling Focus

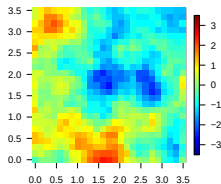
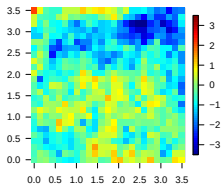
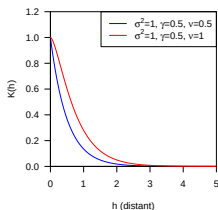
Spatial statisticians often focus on specifying and estimating the covariance function $C(h)$.

Matérn Covariance Function

$$C(h) = \sigma^2 \frac{(\sqrt{2\nu}h/\rho)^\nu K_\nu(\sqrt{2\nu}h/\rho)}{\Gamma(\nu)2^{\nu-1}}, \quad \sigma^2 > 0, \rho > 0, \nu > 0,$$

where

- ν : smoothness parameter
- $K_\nu(\cdot)$: modified Bessel function of the second kind



Next, we will discuss covariance parameter estimation, beginning with **variogram**-based methods and then progressing to modern **likelihood**-based approaches.

Variogram, Semivariogram, and Covariance

For two spatial locations s_i and s_j , the **semivariogram** is defined as

$$\gamma(s_i, s_j) = \frac{1}{2} \text{Var}\{Y(s_i) - Y(s_j)\}.$$

Under second-order stationarity,

$$E\{Y(s)\} = \mu, \quad \text{Cov}\{Y(s_i), Y(s_j)\} = C(s_i - s_j).$$

Therefore,

$$2\gamma(s_i, s_j) = E\left[\{Y(s_i) - Y(s_j)\}^2\right]$$

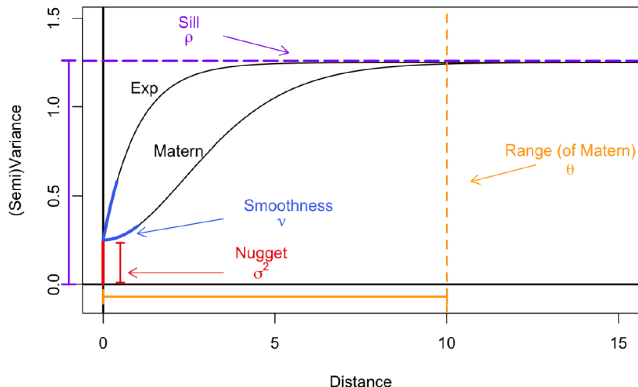
If the process is also isotropic, then the semivariogram depends only on distance $h = \|s_i - s_j\|$, and

$$\gamma(h) = C(0) - C(h)$$

Interpretation

Small semivariogram values indicate strong similarity between nearby locations

Semivariogram $\left\{ \frac{1}{2} \text{Var} (\varepsilon(s_i) - \varepsilon(s_j)) \right\}_{i,j}$



Source: `fields` vignette by Wiens and Krock, 2019

- **Nugget:** error / microscale variation
- **Sill = Nugget + Partial Sill:** total variance
- **Range:** dependence scale
- **Smoothness:** process roughness

Key idea: The empirical semivariogram guides covariance model selection

Estimation: Weighted Least Squares

A common approach is to fit a parametric semivariogram model

$$\gamma(h; \boldsymbol{\theta})$$

to the empirical semivariogram

$$\hat{\gamma}(h_u)$$

The weighted least squares estimator solves

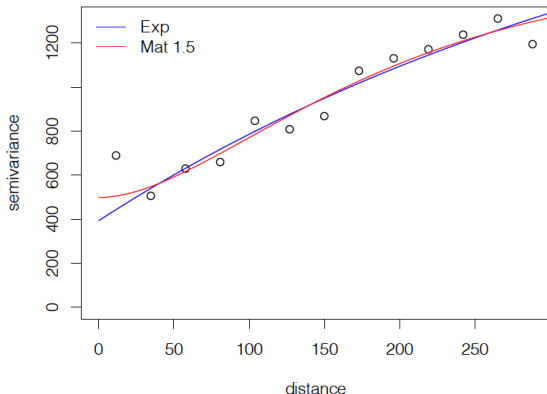
$$\arg \min_{\boldsymbol{\theta}} \sum_{u \in U} \frac{N(h_u)}{\{\gamma(h_u; \boldsymbol{\theta})\}^2} [\hat{\gamma}(h_u) - \gamma(h_u; \boldsymbol{\theta})]^2$$

Why These Weights?

The empirical semivariogram is more precise when:

- more location pairs are available ($N(h_u)$ is large);
- the semivariogram value is small, since the variability of $\hat{\gamma}(h_u)$ typically increases with $\gamma(h_u)^2$.

Fitting Semivariogram Models



Takeaway

Fitted semivariogram models provide the covariance parameters required for kriging.

Next, we consider maximum likelihood estimation (MLE), which uses the full spatial dependence structure

Suppose

$$Y(s) = x(s)^T \beta + \epsilon(s),$$

where $\epsilon(s)$ is a Gaussian process with covariance

$$\text{Cov}\{\epsilon(s_i), \epsilon(s_j)\} = C_\theta(s_i, s_j).$$

For observed locations $s_1, \dots, s_n$,

$$\mathbf{Y} = (Y(s_1), \dots, Y(s_n))^T$$

has a multivariate normal distribution

$$\mathbf{Y} \sim N(X\beta, \Sigma_\theta),$$

where

$$\Sigma_\theta(i, j) = C_\theta(s_i, s_j).$$

MLE estimates β and θ by maximizing the multivariate normal likelihood

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For a fixed covariance parameter vector θ , the generalized least squares estimator is

$$\hat{\beta}(\theta) = (X^T \Sigma_{\theta}^{-1} X)^{-1} X^T \Sigma_{\theta}^{-1} \mathbf{y}.$$

Substituting $\hat{\beta}(\theta)$ into the likelihood gives the profile log-likelihood:

$$\ell(\theta; \mathbf{y}) \propto -\frac{1}{2} \log |\Sigma_{\theta}| - \frac{1}{2} \mathbf{y}^T P(\theta) \mathbf{y},$$

where

$$P(\theta) = \Sigma_{\theta}^{-1} - \Sigma_{\theta}^{-1} X (X^T \Sigma_{\theta}^{-1} X)^{-1} X^T \Sigma_{\theta}^{-1}.$$

Goal

Maximize the profile likelihood to estimate the covariance parameters.

Remarks on Likelihood-Based Estimation

Likelihood-based estimation requires solving a constrained nonlinear optimization problem

- Numerical optimization is usually required
- REML can be used as an alternative to ML
- Large spatial datasets create computational challenges

Computational Cost

Likelihood evaluation typically requires

$$O(n^3)$$

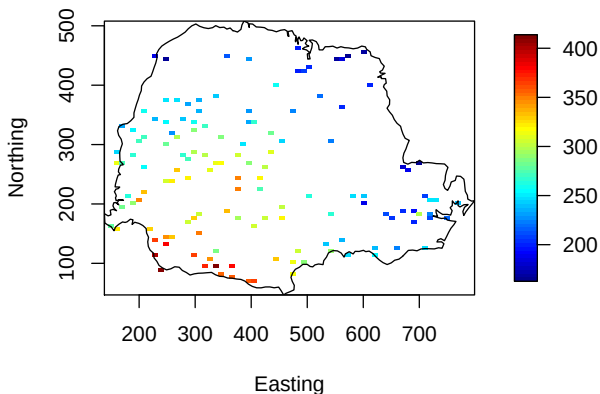
operations and

$$O(n^2)$$

memory because of covariance matrix factorization and storage

Paraná State Precipitation Data

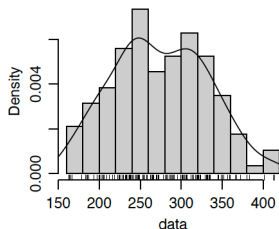
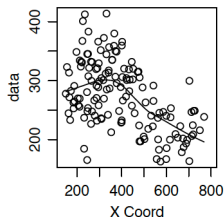
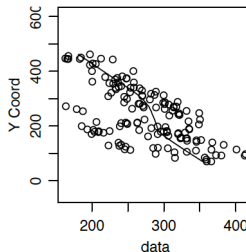
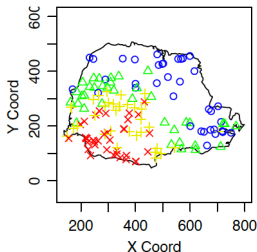
We analyze average winter rainfall during the dry season, May–June, at 143 locations in Paraná, Brazil



Goal

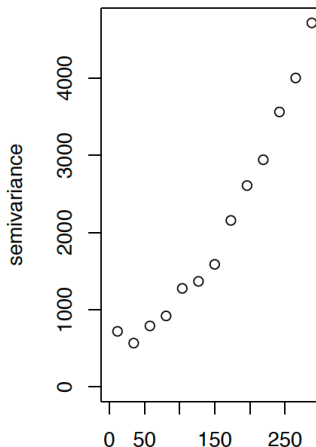
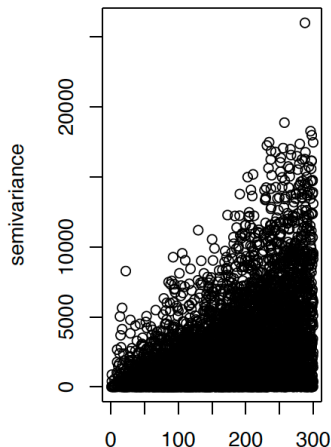
Interpolate precipitation over the spatial domain and quantify prediction uncertainty

Exploratory Data Analysis



A linear trend in space (both longitude and latitude) may be suitable to characterize the large-scale spatial trend

Variogram Analysis

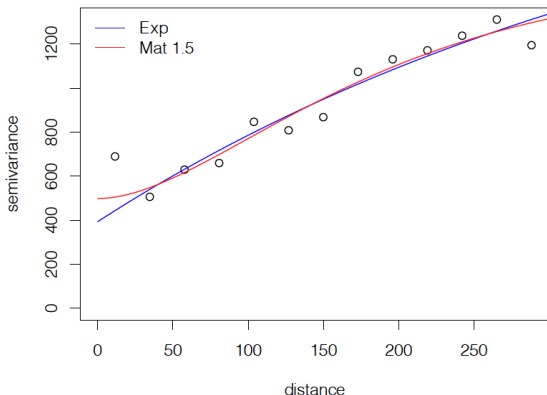


Interpretation

Nearby locations have more similar precipitation values than distant locations, suggesting positive spatial dependence

Estimating Spatial Covariance Via Varogram

```
parana.vtfit.exp <- variofit(parana.variot)  
parana.vtfit.mat1.5 <- variofit(parana.variot, kappa = 1.5)
```



We fit candidate semivariogram models to the empirical variogram

The fitted variogram provides initial estimates of covariance parameters for maximum likelihood estimation

Maximum Likelihood Estimation of Paraná Rainfall

```
(parana.ml1 <- likfit(parana, trend = "1st", ini = c(1000, 50), nug = 100))
```

```
## -----  
## likfit: likelihood maximisation using the function optim.  
## likfit: Use control() to pass additional  
##         arguments for the maximisation function.  
##         For further details see documentation for optim.  
## likfit: It is highly advisable to run this function several  
##         times with different initial values for the parameters.  
## likfit: WARNING: This step can be time demanding!  
## -----  
## likfit: end of numerical maximisation.  
  
## likfit: estimated model parameters:  
##      beta0      beta1      beta2      tausq      sigmasq      phi  
## "416.4984" " -0.1375" " -0.3997" "385.5180" "785.6904" "184.3863"  
## Practical Range with cor=0.05 for asymptotic range: 552.3719  
##  
## likfit: maximised log-likelihood = -663.9
```

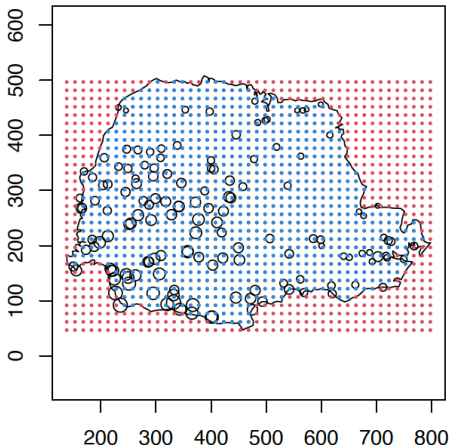
- The mean model includes a first-order spatial trend
- The covariance model captures residual spatial dependence

Next Step

Use the estimated mean and covariance parameters to perform spatial interpolation.

Setting Up the Spatial Grids for Prediction

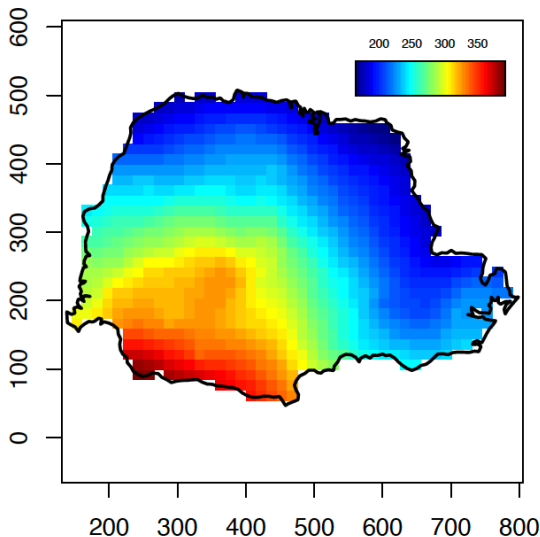
To create a prediction map, we define a grid of prediction locations over the spatial domain



Practical Step

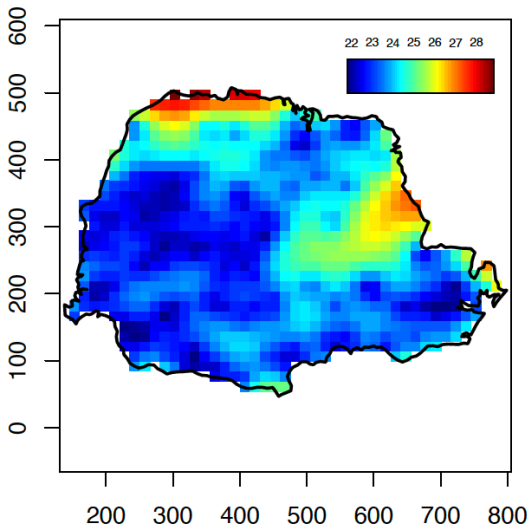
Kriging is applied at each grid point to produce a spatial prediction surface

Spatial Predicted Map



The smooth spatial surface reflects both the fitted mean trend and the estimated spatial covariance structure

Prediction Uncertainty Map



Prediction uncertainty is generally smaller near observed monitoring stations and larger in regions with sparse observations or near the boundary

These slides cover:

- Parameter Estimation for **Gaussian Process Spatial Models**
- **Spatial predictions** using Gaussian Process Spatial Models

R functions to know:

- `quilt.plot` (under the package `fields`) for visualizing irregularly distributed spatial data
- `vgram` and `variog` (under the package `fields` and `geoR`, respectively) for visualizing spatial dependence
- `variofit` and `likfit` from the package `geoR` for conducting **weighted least squares** and **maximum likelihood estimation**

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